DASC 3230 Individual Project 1

Predicting Concrete Strength for Unseen Recipes

Individual assessment · 100 marks

Predict concrete strength for recipes not seen during training. Compare polynomial OLS, ridge, and lasso using the supplied data. Select by cross-validation, evaluate on reserved test data, and produce reusable predictions.

Complete the notebook's **16 TODOs**, three split-code answers **S1-S3**, two standardization answers **S4-S5**, the **baseline prediction answer**, and four result responses **R1-R4**. Implement the modelling workflow using the supplied function interfaces. Keep the checks and answer the short interpretation questions.

Data and tools

The UCI Concrete Compressive Strength dataset contains 1,030 observations before duplicate removal. Preserve this input order:

Inputs	Unit
1. Cement; 2. Blast furnace slag; 3. Fly ash; 4. Water	kg/m³
5. Superplasticizer; 6. Coarse aggregate; 7. Fine aggregate	kg/m³
8. Age	days
Response: concrete compressive strength	MPa

Use **NumPy**, the **Python standard library**, and **Matplotlib for plotting only**. From scikit-learn, you may use only `PolynomialFeatures` and `Lasso`. Do not use other scikit-learn functions, pandas, SciPy, statsmodels, or other modelling libraries. **Do not use PyTorch, TensorFlow, or JAX.**

Assessment requirements

Keep the supplied `SEED = 3230` split and five CV folds. Keep recipe groups separate across development/test and training/validation boundaries. Do not inspect test responses during model development. Complete **R1 before test evaluation**.

Implement the specified preprocessing structure: raw-input standardization, polynomial expansion, and expanded-feature standardization. Fit all preprocessing statistics on the fitting portion only. Retain the course convention `PolynomialFeatures(degree=degree)` with its default bias column; ensure consistent intercept handling across methods.

Evaluate all **21 candidates** on the same folds:

Family	Degrees	λ values
OLS	1, 2, 3	None
Ridge	1, 2, 3	0.1, 10, 1000
Lasso	1, 2, 3	10, 100, 1000

Use the objectives below, with the intercept excluded from both penalties:

$$\text{RSS} = \|y - \Phi\beta\|_2^2, \qquad \text{ridge}: \ \text{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2, \qquad \text{lasso}: \ \text{RSS} + \lambda \sum_{j=1}^{p} |\beta_j|.$$

Implement OLS and ridge with NumPy. Match your Lasso call to the stated objective. In `fit_ols`, `fit_ridge`, and `fit_lasso`, return only the coefficient vector, including the intercept. In `fit_final`, return the fitted feature state with that vector stored under `beta`. Retain the supplied solver settings and resolve any failed checks or convergence errors before evaluation. Use ordinary loops, indexing, and NumPy arrays, as in the course notebooks.

Implement the candidate-by-fold CV loop. Store training and validation MSE in arrays of shape `(21, 5)`, with candidates in rows and folds in columns. Select by the **equally weighted mean of five fold MSEs**. Report fold variation and CV RMSE: the square root of mean fold MSE. Compare with the mean-response and degree-1 OLS baselines.

After selection, implement the final refit on all development observations. Evaluate only the selected model and the pre-specified degree-1 OLS reference on test data, in one fixed comparison. Do not revise the selection using test results. When predicting with your saved model, reuse its degree, scaling statistics, and coefficients without re-estimating them.

Work plan

Part	Required work
A	Remove exact duplicates; audit data; answer S1-S3
B	Implement standardization and MSE
C	Fit the complete feature transformation and OLS; answer S4-S5
D	Implement ridge and lasso
E	Implement CV and selection; plot OLS errors; inspect the supplied family comparison; answer baseline and R1
F	Implement refitting, prediction, test comparison, and diagnostic plots; answer R2-R3
G	Run supplied export checks; inspect queries; answer R4

Use the supplied code for group splitting, candidate settings, plot layout, and file export. Implement the statistical transformations, fitting, CV, selection, final evaluation, and plot data. Use the supplied second CV figure to compare OLS, ridge, and lasso at each degree; for ridge and lasso, each point uses the best supplied lambda for that degree.

Deliverables and marking

Submit **`executed.ipynb`**. Ensure your notebook runs from a fresh kernel and includes all outputs, figures, S1-S5, the baseline prediction answer, and R1-R4. Verify that loading the saved model reproduces the same predictions. Preserve query identifiers and order in the prediction file.

Component	Marks
Data audit and group separation, including S1-S3	8
Standardization and feature preparation, including S4-S5	20
OLS and the degree-error plot	6
Ridge and lasso	14
CV, model selection, and the baseline prediction answer	24
Final refit, reusable prediction, and test evaluation plots	14
Interpretation of results: R1 (4), R2 (2), R3 (4), R4 (4)	14
Total	**100**

Attribution: Yeh, I. (1998). *Concrete Compressive Strength* [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C5PK67, CC BY 4.0. File provenance: `data/README.pdf`.